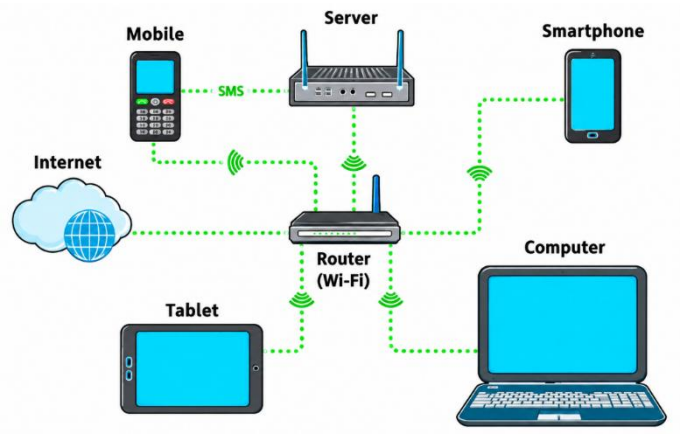
Section 1

Networking basics — LAN, WAN, routers, switches

Networking: Networking is the **process of connecting computers, devices**, or people so they can communicate and share information or resources (Data).

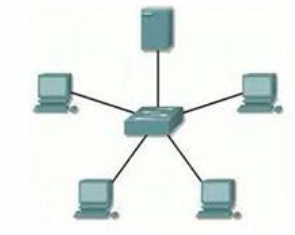
Why we use Network

- Share info or data
- Share Hardware and Software
- Emails or message
- Audio and Video calls
- Share printers
- Share Hard drives



- Switch: add more intelligence

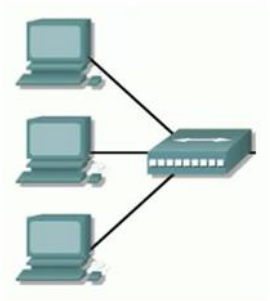
Example of switch



Switch talks to a different devices across your local area network. Sends your request to a relevant network device.

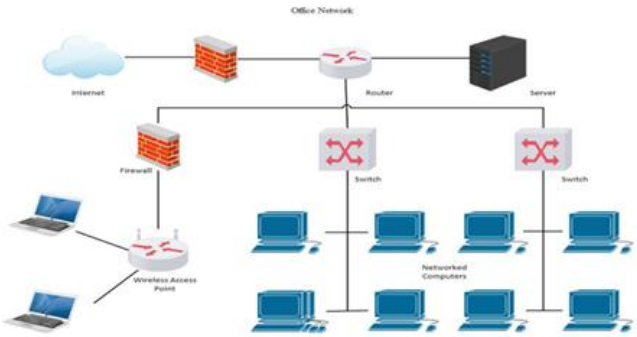
- **Hub**: Connects group of hosts

Example of Hub

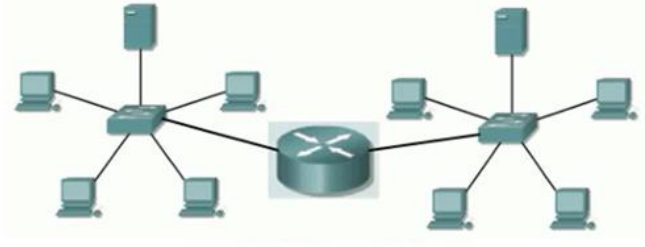




- **Router:** connects different networks or distance networks



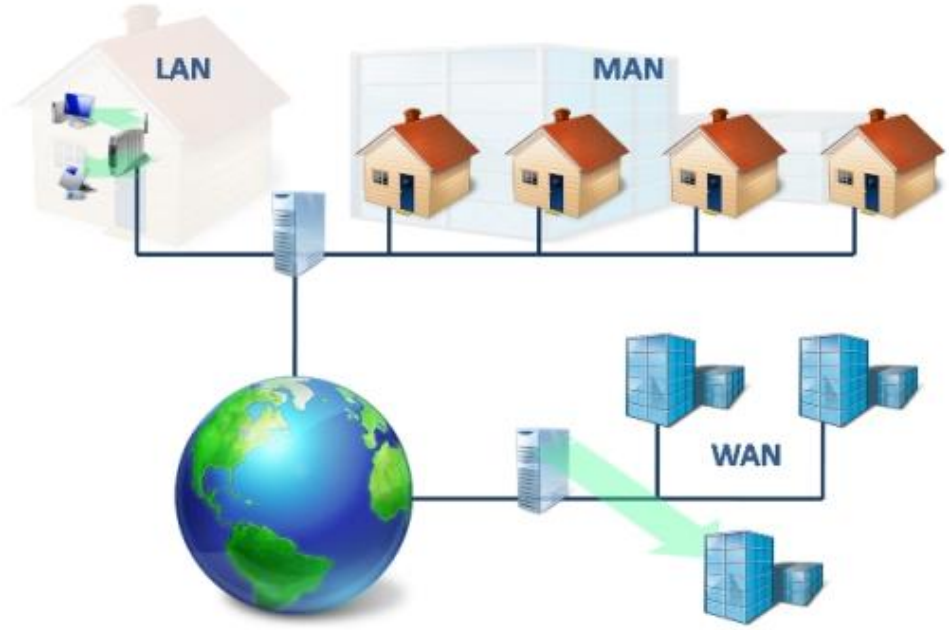
Example of Router



Local Area Networks (LANs) -> Operate within a limited geographic area such as office or Home

Metropolitan-area Networks (MANs) ->

Wide-area Networks (WANs) -> Operate over a large geographical area or connect LANs.



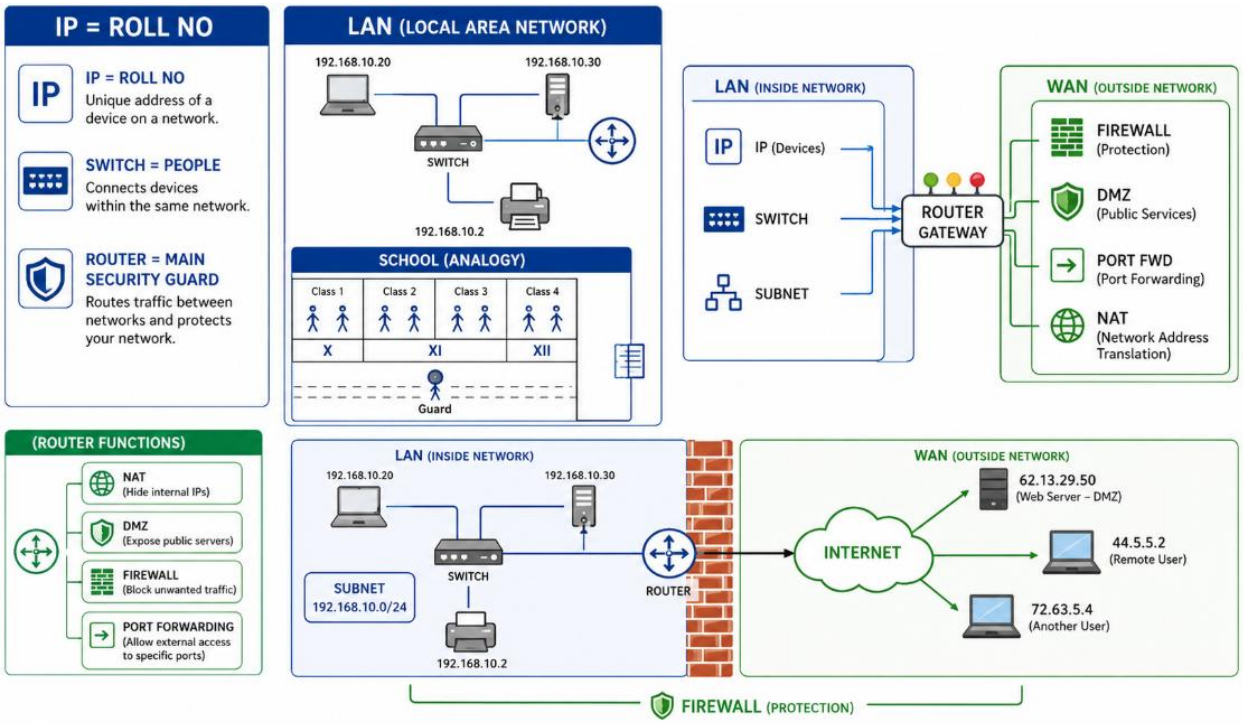
Bandwidth:

Speed of internet is called Bandwidth.

Firewall



Firewall has DMZ (Demilitarized Zone ad-hock or direct connection will be reported to DMZ) and Port FWD (identifies port for our local introduced (subnet) devices).

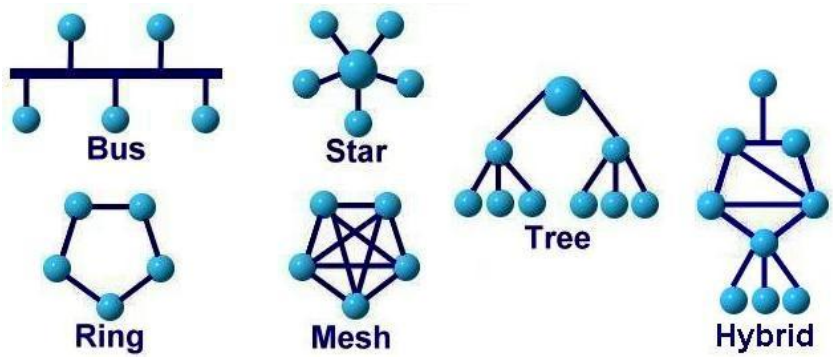


Measuring Bandwidth

Bits per second	Abbreviation	Equivalence
Bits Per Second	bps	1 bps=fundamental unit of bandwidth
Kilobits Per Second	kbps	1kpbs=~1,000 bps=10^3 bps
Megabits Per Second	Mbps	1 Mbps=~1,000,000 bps =10^6 bps
Gigabits Per Second	Gbps	1 Gbps=~1,000,000,000 bps =10^9 bps
Terabits Per Second	Tbps	1 Tbps=~1,000,000,000,000 bps= 10^{12} bps

Shape of Possible Networks:

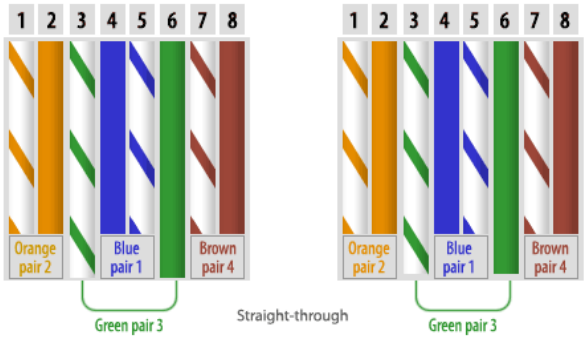
Mainly used are **star** and **tree**



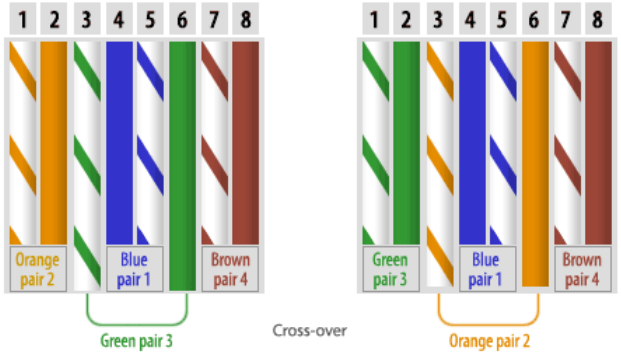
Cable Types

Cat 5 (Category 5) and **Cat 5e (Category 5 enhanced)** are Ethernet cables that use twisted pairs of copper

- **Straight through**: connects two different type of devices such as computer to switch, Router to switch. A straight-through cable has the same wiring standard on both ends**.** (**RJ45** connector)



- **Cross over** : connects two like network devices



Protocols- of transport layer

- At the **Transport Layer**, you can find these common protocols:
- **SPX** – Sequenced Packet Exchange
- **TCP** – Transmission Control Protocol
- **UDP** – User Datagram Protocol
- **SCTP** – Stream Control Transmission Protocol

Protocols – of Network Layer

- At The Network Layer You Can Find These Common Protocols:
- ATP – Apple Talk Transaction Protocol
- IPV4 – Internet Protocol V4
- Ipv6 – Internet Protocol
- IPX – Internetwork Packet Exchange
- ICMP – Internet Control Message Protocol
- IGMP – Internet Group Management Protocol
- OSPF – Open Shortest Path First

IP addressing — IPv4, IPv6, Subneting

Any network needs 3 things IP, Subnet and Gateway (IP address of Router is called gateway)

LAN Card has two addresses as below

✓ IP Address
✓ MAC Address

IP (internet protocol) Address:

MAC (Media access control) Address

A changeable logical address assigned to acms computer or device to identify it and communicate with other devices on a network.

✓ once cabling id done, we must assign IP address for all the network computers. After that the network computers can communicate, share, apply policy, and sending message etc. to others.

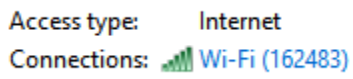
CMD = > getmac to get current device mac address (ipconfig)
The TCP/IP suite defined five blocks of addresses, called **IP address classes**, for specific network uses and sizes.

Address Class	Range	Default Subnet Mask	Networks/Nodes
Class A	1.0.0.0 to 127.255.255.255	255.0.0.0	126 networks of up to 16,777,214 nodes each
Class B	128.0.0.0 to 191.255.255.255	255.255.0.0	16,382 networks of up to 65,534 nodes each
Class C	192.0.0.0 to 223.255.255.255	255.255.255.0	2,097,150 networks of up to 254 nodes each
Class D	224.0.0.0 to 239.255.255.255	None	All members of the multicast session share the same IP address
Class E	240.0.0.0 to 255.255.255.255	None	Strictly for research and experimentation purposes

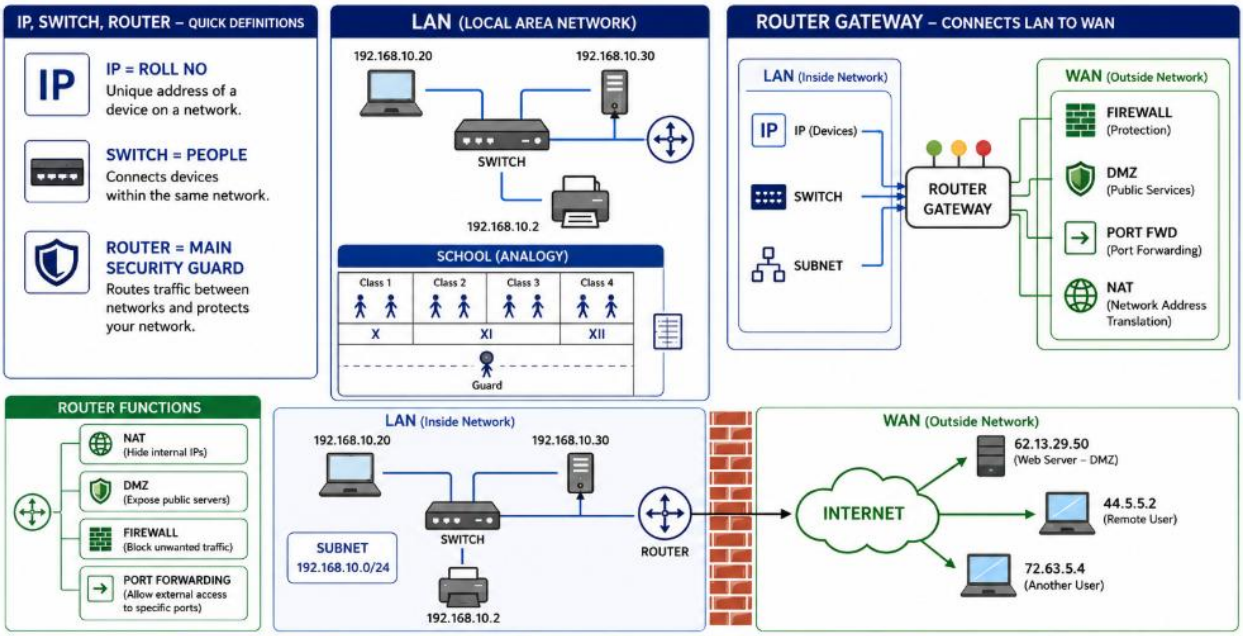
Ex: IP 192.168.1.0
Ex: Sub netmask 255.255.255.0

Means we have 3 octet of Networks and 1 Host. So N.N.N.H so we can't change Networks but we can change Host and also we can do sub-netting.

If you want assign new IP address:
Control Panel, Network and Internet – Network and Sharing Center- Properties – TCP/ip4



IPv4 has totally 32 – bit logical address





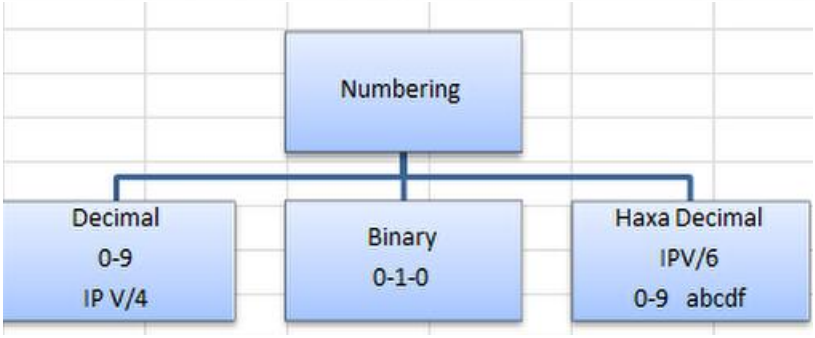
5 IP CLASSES → A | B | C | D | E

Class	Range (First Octet)	Default Subnet Mask	Network/Host Bits	Usable Hosts per Network
A	0 – 126	255.0.0.0	8 / 24	16,777,214
B	128 – 191	255.255.0.0	16 / 16	65,534
C	192 – 223	255.255.255.0	24 / 8	254
D	224 – 239	N/A (Multicast)	N/A	N/A
E	240 – 255	N/A (Experimental)	N/A	N/A

Legend for Class Table
N = Network Bit
H = Host Bit

N	N	N	H	H
N	N	N	H	H
N	H	H	H	H

Note:
Class A, B, C are used for normal networks.



Subnet	1	2	4	8	16	32	64	128	256
Host	256	128	64	32	16	8	4	2	1
Subnet Mask	/24	/25	/26	/27	/28	/29	/30	/31	/32
	0	128 64 192	192 32 224	224 16 240	240 8 248	248 4 252			

192.168.4.0/24.

Make 3 subnets

Total subnets 4
Total host IDs 64 - 2 = 62 including network ID and broadcast ID
Subnet Mask 255.255.255.26

205.150.65.0/26
EX 205.150.65.0/26
SM 255.255.255.192
N-ID 205.150.65.0
B-ID 205.150.65.63
Host Range ID
205.150.65.1-205.150.65.62

Network ID	Subnet Mask	Host ID Range	# of Usable Host	Broadcast ID
192.168.4.0	/26	192.168.4.1-192.168.4.62	64- 2 = 62	192.168.4.63
0+64 = 64 192.168.4.64	/26	192.168.4.65-192.168.4.126	62	192.168.4.127
64+64 = 128 192.168.4.128	/26	192.168.4.129-192.168.4.190	62	192.168.4.191
128+64 = 192 192.168.4.192	/26	192.168.4.193-192.168.4.254	62	192.168.4.255

Subnet defines your local area network Range. Subletting are private IP address that defines some one's computer.
All classes are divided into two Parts Private IPs and Public IPs

ICANN has set aside three nonroutable address ranges that a company can use internally to enable their network nodes to communicate with one another using TCP/IP.
The private, nonroutable IP address ranges are:

10.0.0.0	to	10.255.255.255
172.16.0.0	to	172.31.255.255
192.168.0.0	to	192.168.255.255

Custom subnet masks topic

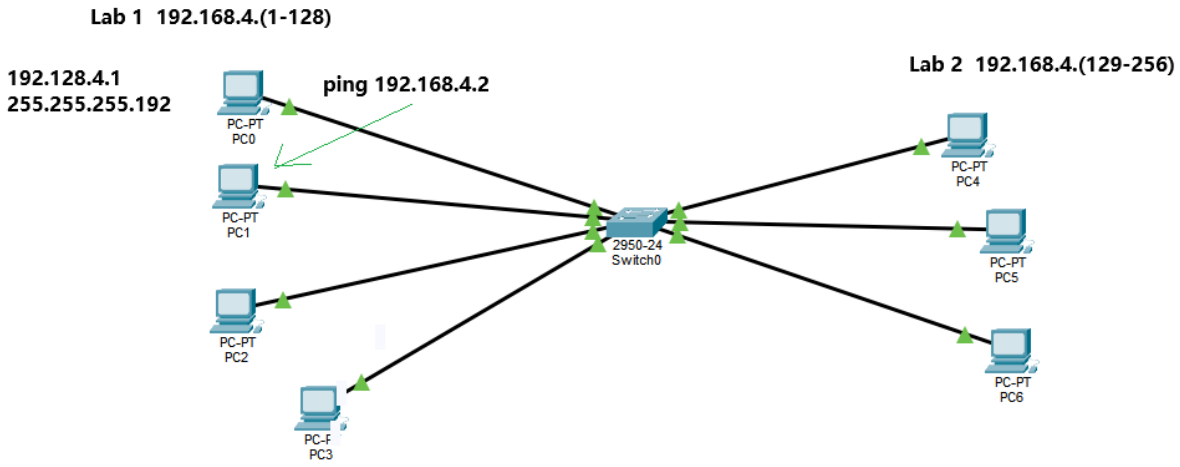
128 16 0 1 128 31 0 1

We can make 15 networks in B class
222.0.0.1 This public IP other than the range of Private IP.

Cisco

Cisco Systems was founded in **1984** by married couple **Leonard Bosack and Sandy Lerner**, who worked at **Stanford University**. The first thing they created was gateway server to communicate. They helped develop networking technology that allowed computers on **different networks and departments to communicate**, which led to the creation of **Cisco routers**. The company was named **Cisco**, derived from **San Francisco**, California. **Cisco Packet Tracer**: is a virtual networking lab used to practice Cisco networking and CCNA skills.

Ex: 8 bits C class IP subnetting 192.168.4.0
192 sub netting



Section 2

Switching — VLANs, trunking, STP

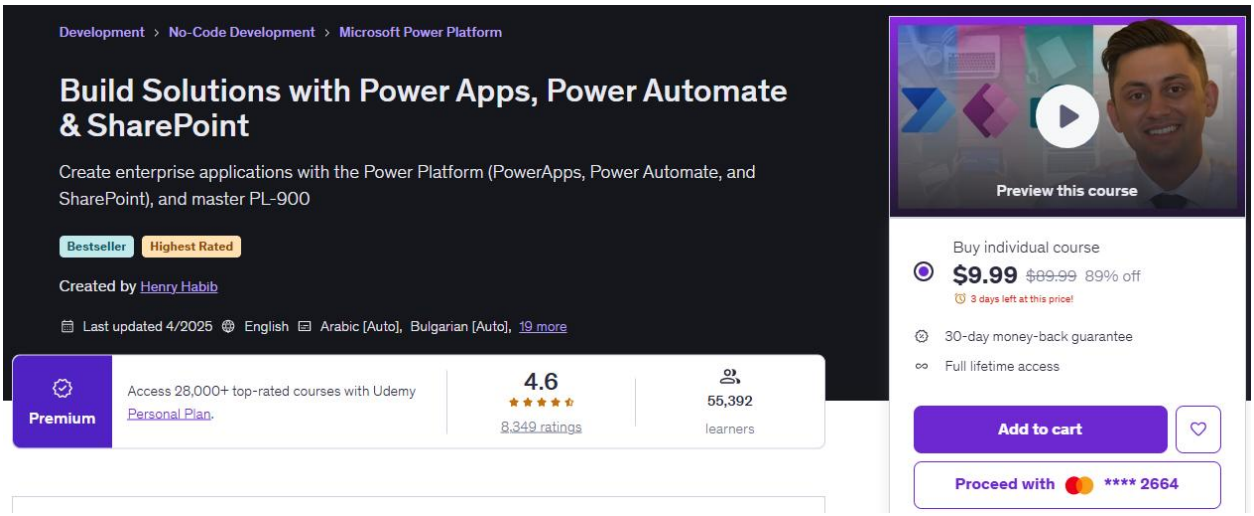
Routing — static routes and OSPF

Wireless networking

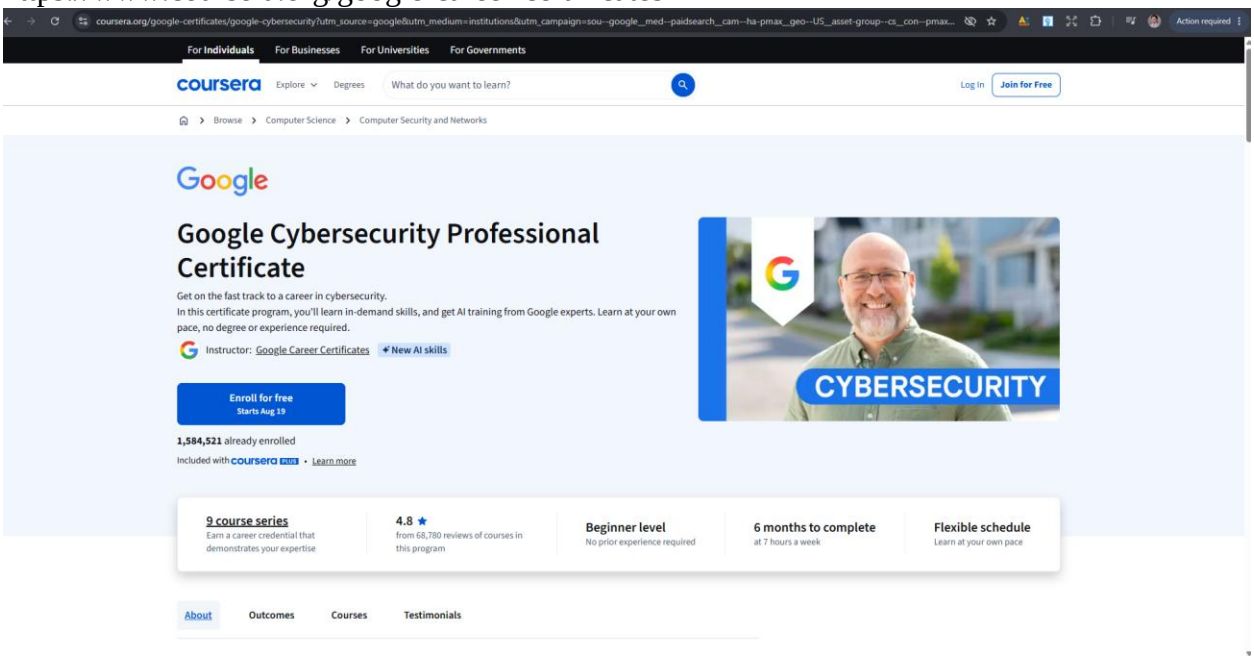
Network security

NAT, DHCP, DNS

Network automation and programmability



https://www.coursera.org/google-career-certificates



https://www.coursera.org/google-certificates/google-cybersecurity?utm_source=google&utm_medium=institutions&utm_campaign=sou--google__med--paidsearch__cam--ha-pmax__geo--US__asset-group--cs__con--pmax__LP--DR__ter--null&gwg_campaign_id=20906443665&gwg_exp=null&gwg_ad_id=GCLID--CjwKCAjwqJXUBhBNEiwA8BgG7vYxmpYotdUV-juPKqYwcXDwTAiRCuJm-j6g_oQjXUZ1ESUE0fEDPBoCGpkQAvD_BwE&_gl=1*1nbigtr*_gcl_aw*R0NMLjE3ODcxNzE5MzUuQ2p3S0NBandxSlhVQmhCTkVpd0E4QmdHN3ZZeG1wWW90ZFVWLWp1UEtxWXdjWER3VEFpUkN1Sm0tajZnX29RalhVWjFFU1VFMGZFRFBCb0NHcGtRQXZEX0J3RQ..*_gcl_au*MjA1NTMyMjQyNC4xNzg3MTcxOTM1